11

Ethernet cabling and connectors

Objectives

When you have completed study of this chapter you will be able to:

- Describe the various types of physical transmission media used for local area networks
- Describe the structure of cables
- Examine factors affecting cable performance
- Describe factors affecting selection of cables
- Explain salient features of AUI cables, coaxial cables, twisted pair cables with their categories, and, fiber optic cables
- Discuss the advantages and disadvantages of each cable type
- Describe Ethernet cabling requirements for various Ethernet media systems
- Describe salient features of various cable connectors
- Describe the use of Ethernet cables and connectors in industrial environments

11.1 Cable types

Three main types of cable are used in networks:

- Coaxial cable, also called coax, which can be thin or thick
- Twisted pair cable, which can be shielded (STP) or unshielded (UTP)
- Fiber optic cables, which can be single-mode, multimode or graded-index multimode

There is also a fourth group of cables, known as IBM cable, which is essentially twisted pair cable, but designed to somewhat more stringent specifications by IBM. Several types are defined, and they are used primarily in IBM token ring networks.

11.2 Cable structure

All cable types have the following components in common:

- One or more conductors to provide a medium for the signal. The conductor might be a copper wire or glass
- Insulation of some sort around the conductors to help keep the signal in and interference out
- An outer sheath, or jacket, to encase the cable elements. The sheath keeps the cable components together, and may also help protect the cable components from water, pressure, or other types of damage

11.2.1 Conductor

For copper cable, the conductor is known as the signal, or carrier, wire, and it may consist of either solid or stranded wire. Solid wire is a single thick strand of conductive material, usually copper. Stranded wire consists of many thin strands of conductive material wound tightly together.

The signal wire is described in the following terms:

- The wire's conductive material (for example, copper)
- Whether the wire is stranded or solid
- The carrier wire's diameter, expressed directly in units of measurement (for example, in inches, centimeters, or millimeters), or in terms of the wire's gauge, as specified in the AWG (American Wire Gauge)
- The total diameter of the strand, which determines some of the wire's electrical properties, such as resistance and impedance. These properties, in turn, help determine the wire's performance
- For fiber optic cable, the conductor is known as the core. The core can be made from either glass or plastic, and is essentially a cylinder that runs through the cable. The diameter of this core is expressed in microns (millionths of a meter)

11.2.2 Insulation

The insulating layer keeps the transmission medium's signal from escaping and also helps to protect the signal from outside interference. For copper wires, the insulation is usually made of a dielectric such as polyethylene. Some types of coaxial cable have multiple protective layers around the signal wire. The size of the insulating layer determines the spacing between the conductors in a cable and therefore its capacitance and impedance.

For fiber optic cable, the 'insulation' is known as cladding and is made of material with a lower refractive index than the core's material. The refractive index is a measure that indicates the manner in which a material will reflect light rays. The lower refractive index ensures that light bounces back off the cladding and remains in the core.

Cable sheath

The outer casing, or sheath, of the cable provides a shell that keeps the cable's elements together. The sheath differs for indoor and outdoor exposure. Outdoor cable sheaths tend to be black, with appropriate resistance to UV light, and have enhanced water resistance. Two main indoor classes of sheath are plenum and non-plenum.

Plenum cable sheath

For certain environments, law requires plenum cable. It must be used when the cable is being run 'naked' (without being put in a conduit) inside walls, and should probably be used whenever possible. Plenum sheaths are made of non-flammable fluoro-polymers such as Teflon or Kynar. They are fire-resistant and do not give off toxic fumes when burning. They are also considerably more expensive (by a factor of 1.5 to 3) than cables with non-plenum sheaths. Studies have shown that cables with plenum sheaths have less signal loss than non-plenum cables. Plenum cable specified for networks installed in the United States should generally meet the National Electrical Code (NEC) CMP (communications plenum cable) or CL2P (class 2 plenum cable) specifications. Networks installed in other countries may have to meet equivalent safety standards, and these should be determined before installation. The cable should also be Underwriters Laboratories (UL) listed for UL-910, which subjects plenum cable to a flammability test.

Non-plenum cable sheath

Non-plenum cable uses less expensive material for sheaths, so it is consequently less expensive than cable with plenum sheaths, but it can often be used only under restricted conditions. Non-plenum cable sheaths are made of polyethylene (PE) or polyvinyl chloride (PVC), which will burn and give off toxic fumes. PVC cable used for networks should meet the NEC CMR (communications riser cable) or CL2R (class 2 riser cable) specifications. The cable should also be UL-listed for UL-1666, which subjects riser cable to a flammability test.

Cable packaging

Cables can be packaged in different ways, depending on what it is being used for and where it is located. For example, the IBM cable topology specifies a flat cable for use under carpets.

The following types of cable packaging are available:

- Simplex cable – one cable within one sheath, which is the default configuration. The term is used mainly for fiber optic cable to indicate that the sheath contains only a single fiber.
- Duplex cable – two cables, or fibers, within a single sheath. In fiber optic cable, this is a common arrangement. One fiber is used to transmit in each direction.
- Multi-fiber cable – multiple cables, or fibers, within a single sheath. For fiber optic cable, a single sheath may contain thousands of fibers. For electrical cable, the sheath will contain at most a few dozen cables.

11.3 Factors affecting cable performance

Copper cables are good media for signal transfer, but they are not perfect. Ideally, the signal at the end of a length of cable should be the same as at the beginning. Unfortunately, this will not be true in practical cables. All signals degrade when transmitted over a distance through any medium. This is because its amplitude decreases as the medium resists the flow of energy, and signals can become distorted because the shape of the electrical signal changes over distance. Any transmission also consists of signal and noise components. Signal quality degrades for several reasons, including attenuation, crosstalk, and impedance mismatches.

11.3.1 Attenuation

Attenuation is the decrease in signal strength, measured in decibels (dB) per unit length. Such loss happens as the signal travels along the wire. Attenuation occurs more quickly at higher frequencies and when the cable's resistance is higher. In networking environments, repeaters are responsible for regenerating a signal before passing it on. Many devices are repeaters without explicitly saying so. Since attenuation is sensitive to frequency, some situations require the use of equalizers to boost signals of different frequencies with the appropriate amount.

11.3.2 Characteristic impedance

The *impedance* of a cable is defined as the resistance offered to the flow of electrical current at a particular frequency. The characteristic impedance is the impedance of an infinitely long cable so that the signal never reaches end of the cable, and hence cannot bounce back. The same situation is replicated when a cable is terminated, so that the signal cannot bounce back. So the characteristic impedance is the impedance of a short cable when it is terminated as shown in Figure 11.1. Such a cable then appears electrically to be infinitely long and has no signal reflected from the termination. If one cable is connected to another of differing characteristic impedance, then signals are reflected at their interface. These reflections cause interference with the data signals and they must be avoided by using cables of the same characteristic impedance.

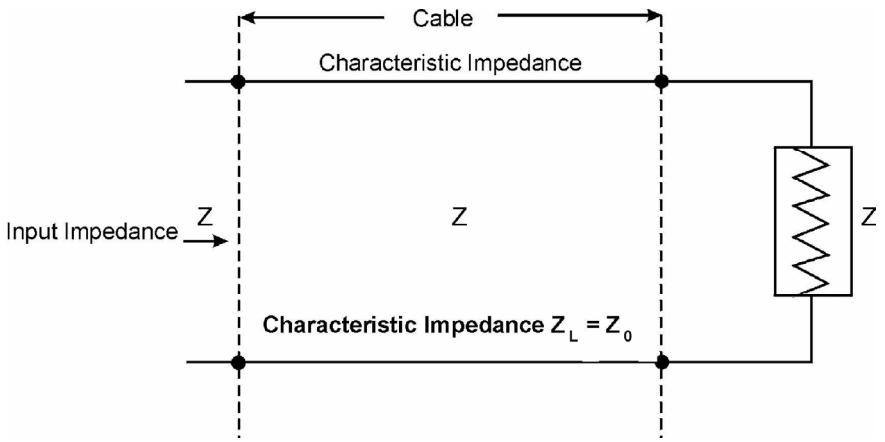


Figure 11.1
Characteristic impedance

11.3.3 Crosstalk

Crosstalk is electrical interference in the form of signals picked up from a neighboring cable or circuits; for example, signals on different wires in a multi-stranded twisted pair cable may interfere with each other. Crosstalk is non-existent in fiber optic cables.

The following forms of cross talk measurement are important for twisted pair cables:

- Near-end cross talk or NEXT: NEXT measurements (in dB) indicate the degree to which unwanted signals are coupled onto adjacent wire pairs. This unwanted 'bleeding over' of a signal from one wire pair to another can distort the desired signal. As the name implies, NEXT is measured at the 'near end' or the end closest to the transmitted signal. NEXT is a 'pair-to-pair' reading where each wire pair is tested for crosstalk relative to another pair. NEXT increases as the frequency of transmission increases. See Figure 11.2.

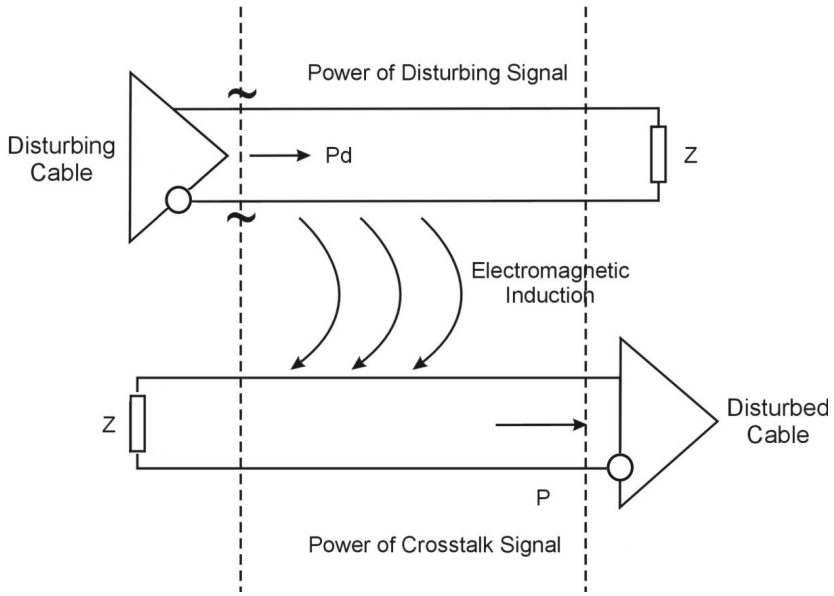


Figure 11.2
Near end crosstalk (NEXT)

- Far-end crosstalk or FEXT is similar in nature to NEXT, but crosstalk is measured at the opposite end from the transmitted signal. FEXT tests are affected by signal attenuation to a much greater degree than NEXT since FEXT is measured at the far end of the cabling link where signal attenuation is greatest. Therefore, FEXT measurements are a more significant indicator of cable performance if attenuation is accounted for
- Equal-level far-end crosstalk or ELFEXT: The comparative measurement of FEXT and attenuation is called equal level far end crosstalk or ELFEXT. ELFEXT is the arithmetic difference between FEXT and attenuation. Characterizing ELFEXT is important for cabling links intended to support 4 pair, full-duplex network transmissions
- Attenuation-to-cross talk ratio or ACR is not specifically a new test, but rather a relative comparison between NEXT and attenuation performance. Expressed in decibels (dB) the ratio is the arithmetic difference between NEXT and attenuation. ACR is significant because it is more indicative of cable performance than NEXT or attenuation alone. ACR is a measure of the strength of a signal compared to the crosstalk noise
- Power sum NEXT – power sum (in dB) is calculated from the six measured pair-to-pair cross talk results. Power Sum NEXT differs from pair-to-pair NEXT by determining the crosstalk induced on a given wire pair from 3 disturbing pairs. This methodology is critical for support of transmissions that utilize all four pairs in the cable such as Gigabit Ethernet. See Figure 11.3.

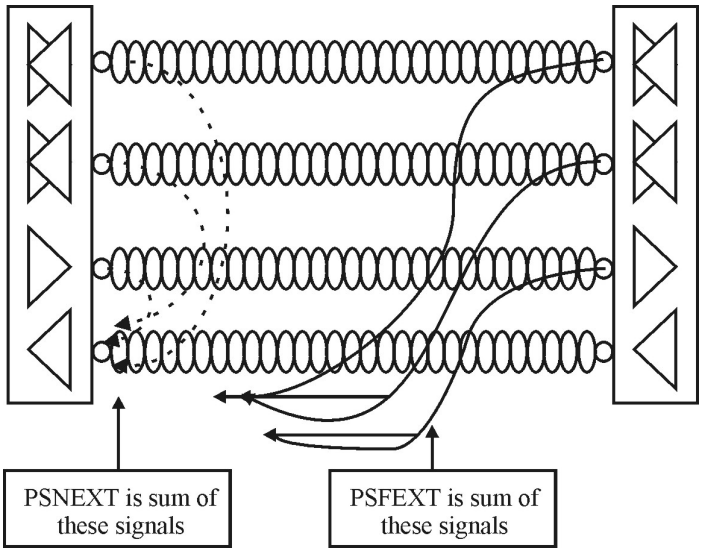


Figure 11.3
PSNEXT and PSFEXT

11.4 Selecting cables

Cables are used to meet all sorts of power and signaling requirements. The demands made on a cable depend on the location in which the cable is used and the function for which the cable is intended. These demands, in turn, determine the features a cable should have.

11.4.1 Function and location

Here are a few examples of considerations involving the cable's function and location:

- Cable designed to run over long distances, such as between floors or buildings, should be robust against environmental factors (moisture, temperature changes, and so on). This may require extra sheaths or sheaths made with a special material. Fiber optic cable performs well, even over distances much longer than a floor or even a building.
- Cable that must run around corners should bend easily, and the cable's properties and performance should not be affected by the bending. For several reasons, twisted pair cable is probably the best cable for such a situation (assuming it makes sense within the rest of the wiring scheme). Of course, another way to get around a corner is by using a connector. However, connectors may introduce signal-loss problems.
- Cables that must run through areas in which heavy current motors are operating (or worse, being turned on and off at random intervals) must be able to withstand magnetic interference. Large currents produce strong magnetic fields, which can interfere with and disrupt nearby signals. Because it is not affected by such electrical or magnetic fluctuations, fiber optic cable is the best choice in machinery-intensive environments.
- If you need to run many cables through a limited area, cable weight can become a factor, particularly if all cables will be running in the ceiling. In general, fiber optic and twisted pair cables tend to be the lightest.
- Cables being installed in barely accessible locations must be particularly reliable. It is worth considering installing a backup cable during the initial

installation. Because the installation costs in such locations are generally much more than the cable material cost, installation costs for the second cable add only marginally to the total cost. Generally, the suggestion is to make at least the second cable optical fiber

- Cables that need to interface with other worlds (for example, with a mainframe network or a different electrical or optical system) may need special properties or adapters. The kinds of cable required will depend on the details of the environments and the transition between them.

11.4.2 Main cable selection factors

Along with the function and location considerations, cable selections are determined by a combination of factors, including the following:

- The type of network being created (for example, Ethernet or token ring) – while it is possible to use just about any type of cable in any type of network, certain cable types have been more closely associated with particular network types.
- The amount of money available for the network –cable installation is a major part of the network costs.
- Cabling resources currently available (and useable) –available wiring that could conceivably be used for a network should be evaluated. It is almost certain, however, that at least some of that wire is defective or is not up to the requirements for the proposed network.
- Building or other safety codes and regulations.

11.5 AUI cable

Attachment unit interface cable (AUI) is a shielded multi-stranded cable used to connect Ethernet devices to Ethernet transceivers, and for no other purpose. AUI cable is made up of four individually shielded pairs of wire surrounded by a shielding double sheath. This shield makes the cable more resistant to signal interference, but increases attenuation over long distance.

Connection to other devices is made through DB15 connectors. Connectors at the end of the cable are male and female respectively. Any cable with male-male or female-female connectors at both ends are non-standard and should not be used.

AUI cable is used to connect transceivers to other Ethernet devices, and transceivers need power to operate. This power may be supplied to transceivers by an external power supply or by a pair of wires in the AUI cable dedicated to power supply.

AUI cable is available in two types, standard AUI and office AUI. Standard AUI cable is made up of 20 or AWG copper wire and can be used for distances up to 50 m. but is 0.420 inch thick and is somewhat inflexible. Office AUI cable is thinner (0.26 inch) and is made up of 28 AWG wire, and is relatively flexible. It can be used over distances of 16.5 m only.

Office AUI cable should be used only when standard version is found to be cumbersome due to its inflexibility.

11.6 Coaxial cables

In coaxial cables, two or more separate materials share a common central axis. Coaxial cables, often called coax, are used for radio frequency and data transmission. The cable is

remarkably stable in terms of its electrical properties at frequencies below 4 GHz, and this makes the cable popular for cable television (CATV) transmissions, as well as for creating local area networks (LANs).

11.6.1 Coaxial cable construction

A coaxial cable consists of the following layers (moving outward from the center) as shown in Figure 11.4.

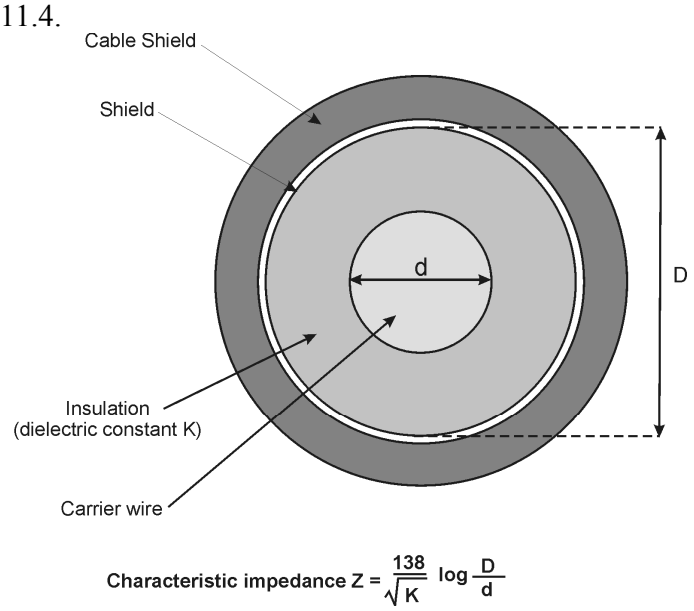


Figure 11.4
Cross-section of a coaxial cable

- **Carrier wire**
  A conductor wire or signal wire is in the center. This wire is usually made of copper and may be solid or stranded. There are restrictions regarding the wire composition for certain network configurations. The diameter of the signal wire is one factor in determining the attenuation of the signal over distance. The number of strands in a multi-strand conductor also affects the attenuation.
- **Insulation**
  An insulation layer consists of a dielectric around the carrier wire. This dielectric is usually made of some form of polyethylene or Teflon.
- **Foil shield**
  This thin foil shield around the dielectric usually consists of aluminum bonded to both sides of a tape. Not all coaxial cables have foil shielding. Some have two foil shield layers, interspersed with copper braid shield layers.
- **Braid shield**
  A braid, or mesh, conductor, made of copper or aluminum, that surrounds the insulation and foil shield. This conductor can serve as the ground for the carrier wire. Together with the insulation and any foil shield, the braid shield protects the carrier wire from electro magnetic interference (EMI) and radio frequency interference (RFI). It should be carefully note that the braid and foil shields provide good protection against electrostatic interference when earthed correctly, but little protection against magnetic interference.
- **Sheath**
  This is the outer cover that can be either plenum or non-plenum, depending on its composition. The layers surrounding the carrier wire also help prevent

signal loss due to radiation from the carrier wire. The signal and shield wires are concentric, or co-axial and hence the name.

11.6.2 Coaxial cable performance

The main features that affect the performance of coaxial cables are its composition, diameter, and impedance:

- The carrier wire's composition determines how good a conductor the cable will be. The IEEE specifies stranded copper carrier wire with tin coating for 'thin' coaxial cable, and solid copper carrier wire for 'thick' coaxial cable. (These terms will be defined shortly.)
- Cable diameter helps determine the electrical demands that can be made on the cable. In general, thick coaxial can support a much higher level of electrical activity than thin coaxial.
- Impedance is a measure of opposition to the flow of alternating current. The properties of the dielectric between the carrier wire and the braid help determine the cable's impedance.
- Impedance determines the cable's electrical properties and limits where the cable can be used. For example, Ethernet and ARCnet architectures can both use thin coaxial cable, but they have different characteristic impedances and so Ethernet and ARCnet cables are not compatible. Most LAN cables have an RG (recommended gauge) rating and cables with the same RG rating from different manufacturers can be safely mixed.

Recommended Gauge	Application	Characteristic impedance
RG-8	10Base5	50 ohms
RG-58	10Base2	50 ohms
RG-59	CATV	75 ohms
RG-2	ARCnet	93 ohms

Table 11.1
Common network coaxial cable impedances

In networks, the characteristic cable impedances range from 50 Ohms (for Ethernet) to 93 Ohms (for ARCnet). The impedance of the coaxial cable in Figure 11.4 is given by the formula:

$Z_0=(138/\sqrt{k})\log(D/d)$ in Ohms

Where k is the dielectric constant of the insulation.

11.6.3 Thick coaxial cable

Thick coaxial (RG-8) cable is 0.5 inch or 12.7 mm in diameter. It is used for 'Thick Ethernet' networks, also called 10Base5 or ThickNet networks. It can also be used for cable TV (CATV), and other connections. Thick coaxial cable is expensive and can be difficult to install and work with. Although 10Base5 is now obsolete, it remains in use in existing installations. The cable is a distinctive yellow, or orange, color, with black stripes every 2.5 meters (8 feet), indicating where node taps can be made.

This cable is constructed with a single solid copper core that carries the network signals, and a series of layers of shielding and insulator material.

Transceivers are connected to the cable at specified distances from one another, and standard transceiver cables connect these transceivers to the network devices.

Extensive shielding makes it highly resistant to electrical interference by outside sources such as lightning, machinery, etc. Bulkiness and limited flexibility of the cable limits its use to backbone media and is placed in cable runways or laid above ceiling tiles to keep it out of the way.

Thick coaxial cable is designed to access as a shared media. Multiple transceivers can be attached to the thick coaxial cable at multiple points on the cable itself. A properly installed length of thick coaxial cable can support up to 100 transceivers.

11.6.4 Thin coaxial cable

Thin coaxial cable (RG-58) is 3/16 inch or 4.76 mm in diameter. When used for IEEE802.3 networks, it is often known as Thin Ethernet. Such networks are also known as 10Base2, 'thinnet', or 'cheapernet'. When using this configuration, drop cables are not allowed. Instead, the T connector is connected directly to the Network Interface Card (NIC) at the node, since the NIC has an on-board transceiver.

It is smaller, lighter, and more flexible than thick coaxial cable. The cable itself resembles (but is not identical to) television coaxial cable.

Thin coaxial cable, due to its less extensive shielding capacity, can be run to a maximum length of 185 meters (606.7 ft).

50-ohm terminators are used on both cable ends.

11.6.5 Coaxial cable designations

Listed below are some of the available coaxial cable types.

- **RG-8:** Used for Thick Ethernet. It has 50 ohms impedance. The Thick Ethernet configuration requires an attachment unit interface (AUI) cable and a media access unit (MAU), or remote transceiver. The AUI cable required is a twisted pair cable that connects to the NIC. RG-8 is also known as N series Ethernet cable.
- **RG-58:** Used for Thin Ethernet. It has 50 ohms impedance and uses a BNC connector.

11.6.6 Advantages of a coaxial cable

A coaxial cable has the following general advantages over other types of cable that might be used for a network.

These advantages may change or disappear over time, as technology advances and products improve:

- The cable is relatively easy to install
- Coaxial cable is reasonably priced compared with other cable types

11.6.7 Disadvantages of coaxial cable

Coaxial cable has the following disadvantages when used for a network:

- It is easily damaged and sometimes difficult to work with, especially in the case of thick coaxial
- Coaxial is more difficult to work with than twisted pair cable
- Thick coaxial cable can be expensive to install, especially if it needs to be pulled through existing cable conduits
- Connectors can be expensive

11.6.8 Coaxial cable faults

The main problems encountered with coaxial cables are:

- Open or short circuited cables (and possible damage to the cable)
- Characteristic impedance mismatches
- Distance specifications being exceeded
- Missing or loose terminator

11.7 Twisted pair cable

Twisted pair cable is widely used, inexpensive, and easy to install. Twisted pair cable comes in two main varieties

- Shielded (STP)
- Unshielded (UTP)

It can transmit data at an acceptable rate – up to 1000 Mbps in some network architectures. The most common twisted pair wiring is telephone cable, which is unshielded and is usually voice-grade, rather than the higher-quality data-grade cable used for networks.

In a twisted pair cable, two conductor wires are wrapped around each other. Twisted pairs are made from two identical insulated conductors, which are twisted together along their length at a specified number of twists per meter, typically forty twists per meter (twelve twists per foot). The wires are twisted to reduce the effect of electromagnetic and electrostatic induction.

For full-duplex digital systems using balanced transmission, two sets of screened twisted pairs are required in one cable; each set with individual and overall screens. A protective PVC sheath then covers the entire cable. (Note: 10BaseT CSMA/CD is not full duplex but still needs 2 pairs)

Twisted pair cables are used with the following Ethernet physical layers

- 10BaseT
- 100BaseTX
- 100BaseT2
- 100BaseT4
- 1000BaseT

The capacitance of a twisted pair is fairly low at about 40 to 160 pF/m, allowing a reasonable bandwidth and an achievable slew rate. A signal is transmitted differentially between the two conductor wires. The current flows in opposite directions in each wire of the active circuit, as shown in Figure 11.5.

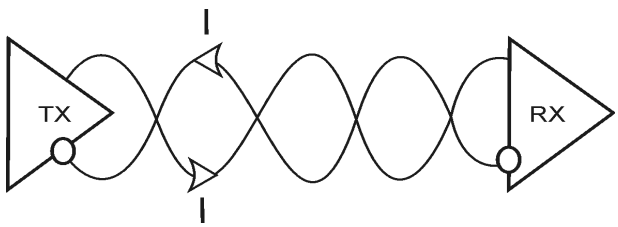


Figure 11.5
Current flow in a twisted pair

11.7.1 Elimination of noise by signal inversion

The twisting of associated pairs and method of transmission reduces the interference of the other strands of wire throughout the cable.

The network signals to transmitted in the form of changes of electrical state. Encoding turns ones and zeroes of network frames into these signals. In a twisted pair system, once a transceiver has been given an encoded signal to transmit, it will invert the polarity of that signal and transmit it on the other wire. The result of this a mirror image of the original signal.

Both the original and the inverted signal are then transmitted over the TX+ and TX– wires respectively. Since these wires are of the same length and have the same construction, the signal travels at the same rate through the cable. Since the pairs are twisted together, any outside electrical interference that affects one member of the pair will have the same effect on both signals.

The transmissions of the original signal and its mirror image reach a destination receiver. This receiver, operating in differential mode, inverts the signal on the TX- line before adding it to the signal on the TX- line. The signal on the TX- line is, however, already inverted so in reality an addition takes place. The noise component on the TX- line, however, is not inverted and as a result it is subtracted from the noise on the TX+ line, resulting in noise cancellation.

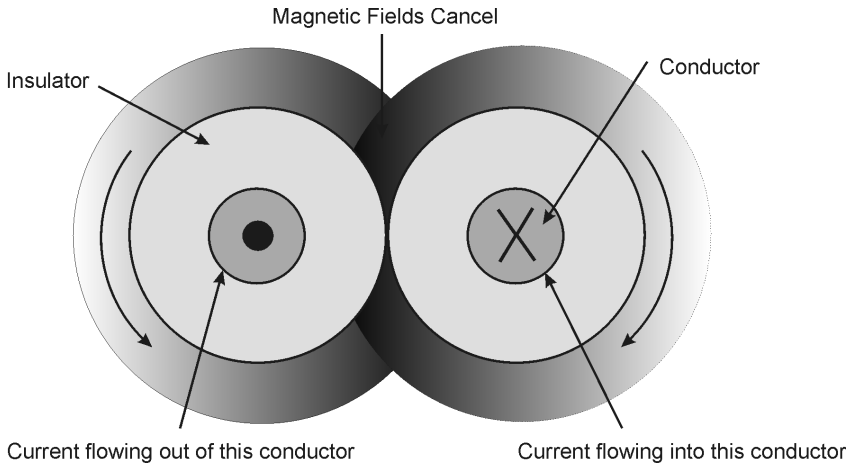


Figure 11.6
Magnetic shielding of twisted pair cables

Since the currents in the two conductors are equal and opposite, their induced magnetic fields also cancel each other. This type of cable is therefore self-shielding and is less prone to interference.

Twisting within a pair minimizes crosstalk between pairs. The twists also help deal with electro magnetic interference (EMI) and radio frequency interference (RFI), as well as balancing the mutual capacitance of the cable pair. The performance of a twisted pair cable can be influenced by changing the number of twists per meter in a wire pair. Each of the pairs in a 4-pair category 5 cable will have a different twist rate to reduce the crosstalk between them.

11.7.2 Components of twisted pair cable

A twisted pair cable has the following components:

- **Conductor wires**
  The signal wires for this cable come in pairs that are wrapped around each other. The conductor wires are usually made of copper. They may be solid (consisting of a single wire) or stranded (consisting of many thin wires wrapped tightly together). A twisted pair cable usually contains multiple twisted pairs; 2, 4, 6, 8, 25, 50, or 100 twisted pair bundles are common. For network applications, 2 and 4 pair cables are most commonly used
- **Shield**
  Some twisted pair cables have a shield in the form of a woven braid. This type of cable is referred to as shielded twisted pair or STP. **Sheath**
  The wire bundles are encased in a sheath made of polyvinyl chloride (PVC) or, in plenum cables, of a fire-resistant material, such as Teflon or Kynar. STP contains an extra shield or protective screen around each of the wire pairs to cut down on extraneous signals. This added protection also makes STP more expensive than UTP.

11.7.3 Shielded twisted pair (STP) cable

STP refers to the 150 ohm twisted pair cabling defined by the IBM cabling specifications for use with token ring networks. 150 ohm STP is not generally used with Ethernet. However, the Ethernet standard does describe how it can be adapted for use with 10BaseT, 100BaseTX, and 100BaseT2 Ethernet by installing special impedance matching transformers, or 'baluns', that convert the 100-ohm impedance of the Ethernet transceivers to the 150-ohm impedance of the STP cable. A balun (BALanced – UNbalanced) is an impedance matching transformer that converts the impedance of one interface to the impedance of the other interface. These are generally used to connect balanced twisted pair cabling with an unbalanced coaxial cabling. These are available from IBM, AMP, and Cambridge Connectors among others.

11.7.4 Unshielded twisted pair (UTP) cable

UTP cable does not include any extra shielding around the wire pairs. This type of cable is used in some slower speed Token Ring networks and can be used in Ethernet and ARCnet systems.

UTP is now the primary choice for many network architectures, with the IEEE approving standards for 10, 100 and 1000 Mbps Ethernet systems using UTP cabling. These are known as:

- 10BaseT for 10 Mbps
- 100BaseTX for 100 Mbps
- 1000 BaseT2 for 1000 Mbps on twisted pair cable

Because it lacks a conductive shield, UTP is not as good at blocking electrostatic noise and interference as STP or coaxial cable. Consequently, UTP cable segments must be shorter than when using other types of cable. For standard UTP, the length of a segment should never exceed 100 meters, or about 330 feet. Conversely, UTP is quite inexpensive, and is very easy to install and work with. The price and ease of installation make UTP tempting, but bear in mind that installation labor is generally the major part of the cabling expense and that other types of cable may be just as easy to install.

Four-pair UTP cable

UTP cabling most commonly includes 4 pairs of wires enclosed in a common sheath. 10BaseT, 100BaseTX, and 100BaseT2 use only two of the four pairs, while 100BaseT4 and 1000BaseT require all four pairs. Two-pair UTP is, however, available and is sometimes used in Industrial Ethernet installations.

The typical UTP cable is a polyvinyl chloride (PVC) or plenum-rated plastic jacket containing four pairs of wire. The majority of facility cabling in current and new installations is of this type. The dedicated (single) connections made using four-pair cable are easier to troubleshoot and replace than the alternative, bulk multi-pair cable such as 25-pair cable.

The insulation of each wire in a four-pair cable will have an overall color: brown, blue, orange, green, or white. In a four-pair UTP cable there is one wire each of brown, blue, green, and orange, and four wires of which the overall color is white. Periodically placed (usually within 1/2 inch of one another) rings of the other four colors distinguish the white wires from one another.

Wires with a unique base color are identified by that base color i.e. “blue”, “brown”, “green”, or “orange”. Those wires that are primarily white are identified as “white/<color>”, where “<color>” indicates the color of the rings.

The 10BaseT and 100BaseTX standards are concerned with the use of two pairs, pair 2 and pair 3 (of either EIA/TIA 568 specification). The A and B specifications are basically the same, except that pair 2 (orange) and pair 3 (green) are swapped. 10BaseT and 100BaseTX configure devices transmit over pair 3 of the EIA/TIA 568A specification (pair 2 of EIA/TIA 568B), and to receive from pair 2 of the EIA/TIA 568A specification (pair 3 of EIA/TIA 568B). The use of the wires of a UTP cable is shown in Table 11.2.

Wire Colour	EIA/TIA Pair	Ethernet Signal Use	
		568A	568Br
White/Blue (W-BL)	Pair 1	Not Used	
Blue (BL)			
White/Orange (W-OR)	Pair 2	RX+	TX+
Orange (OR)		RX-	TX-
White/Green (W-GR)	Pair 3	TX+	RX+
Green (GR)		TX-	RX-
White/Brown (W-BR)	Pair 4	Not Used	
Brown (BR)			

Table 11.2
Four-pair wire use for 10BaseT and 100BaseTX

Twenty-five pair cable

UTP cabling in large installations requiring several cable runs between two points is often 25-pair cable. This is a heavier, thicker form of UTP. The wires within the plastic jacket are of the same construction, and are twisted around associated wires to form pairs, but there are 50 individual wires twisted into 25 pairs in these larger cables. In most cases, 25-pair cable is used to connect wiring closets to one another, or to distribute large amounts of cable to intermediate distribution points, from which four-pair cable is run to the end stations.

Wires within a 25-pair cable are identified by color. The insulation of each wire in a 25-pair cable has an overall color: violet, green, brown, blue, red, orange, yellow, gray, black, and white.

In a 25-pair UTP, cable two colors identify all wires in the cable. The first color is the base color of the insulator; the second color is the color of narrow bands on the base color. These identifying bands are periodically placed on the wire, and repeated at regular intervals. A wire in a 25-pair cable is identified first by its base color, and then further specified by the color of the bands.

As a 25-pair cable can be used to make up to 12 connections between Ethernet stations (two wires in the cable are typically not used), the wire pairs need to be identified not only as transmit/receive pairs, but what other pair they are associated with.

There are two methods of identifying sets of pairs in a 25-pair cable.

The first is based on the connection of a 25-pair cable to a specific type of connector designed especially designed for it, the RJ-21 connector. The second is based on connection to a punch down block, a cable management device typically used to make the transition from a single 25-pair cable to a series of four-pair cables easier.

Crossover connections for 10BaseT and 100BaseTX

Crossing over is the reversal of transmit and receive pairs at opposite ends of a single cable. The 10BaseT and 100BaseTX specifications require that some UTP connections be crossed over.

Those cables that maintain the same pin numbers for transmit and receive pairs at both ends are called straight-through cables.

The 10BaseT and 100BaseTX specifications are designed around connections from the networking hardware to the end user stations being made through straight-through cabling. Thus, the transmit wires of a networking device such as a stand-alone hub or repeater connect to the receive pins of a 10BaseT or 100BaseTX end station.

If two similarly designed network devices, e.g. two hubs, are connected using a straight-through cable, the transmit pins of one device are connected to the transmit pins of the other device. In effect, the two devices will both attempt to transmit on the same pair.

A crossover must therefore be placed between two similar devices, so that transmit pins of one device to connect to the receive pins of the other device. When two similar devices are being connected using UTP cabling, an odd number of crossover cables, preferably only one, must be part of the cabling between them.

Screened twisted pair (ScTP) cables

Screened twisted pair cable, also referred to as foil twisted pair (FTP) is a 4-pair 100-ohm UTP, with a foil screen surrounding all four pairs in order to minimize EMI radiation and susceptibility to outside noise. This is simply a shielded version of the category 3, 4, and 5 UTP cable. It may be used in Ethernet applications in the same manner as equivalent category of UTP cable.

There are versions available where individual screens wrap around each pair.

11.7.5 EIA/TIA 568 cable categories

To distinguish varieties of UTP, the EIA/TIA has formulated several categories. The electrical specifications for these cables are detailed in EIA/TIA 568A, TSB-36, TSB-40 and their successor SP2840.

These categories are:

- **Category 1**
  Voice-grade, UTP telephone cable. This describes the cable that has been used for years in North America for telephone communications. Officially, such cable is not considered suitable for data-grade transmissions. In practice,

however, it works fine over short distances and under ordinary working conditions. It should be known that other national telecommunications providers have often used cable that does not even come up to this minimum standard, and as such, is unacceptable for data transmission.

- **Category 2**
  Voice-grade UTP, although capable of supporting transmission rates of up to 4 Mbps. IBM type 3 cable falls into this category.
- **Category 3**
  Data-grade UTP, used extensively for supporting data transmission rates of up to 10 Mbps. An Ethernet 10BaseT network requires at least this category of cable. Category 3 UTP cabling must not produce an attenuation of a 10 MHz signal greater than 98 dB/km at the control temperature of 20°C. Typically category 3 cable attenuation increases 1.5% per degree Celsius.
- **Category 4**
  Data-grade UTP, capable of supporting transmission rates of up to 16 Mbps. An IBM Token Ring network transmitting at 16 Mbps requires this type of cable. Category 4 UTP cabling must not produce an attenuation of a 10 MHz signal greater than 72 dB/km at the control temperature of 20°C.
- **Category 5**
  Data-grade UTP, capable of supporting transmission rates of up to 155 Mbps (but officially only up to 100 Mbps). Category 5 cable is constructed and insulated such that the maximum attenuation of a 10 MHz signal in a cable run at the control temperature of 20°C is 65 dB/km. TSB-67 contains specifications for the verification of installed UTP cabling links that consist of cables and connecting hardware specified in the TIA-568A standard.
- **Enhanced category 5 standard (category 5e)**
  "Enhanced Cat5" specifies transmission performance that exceeds that of Cat5, and it is used for 10BaseT, 100BaseTX, 155 Mbps ATM, etc. It has improved specifications for NEXT, PSELFEXT and attenuation.
  Category 5e directly supports the needs of Gigabit Ethernet.
  Its frequency range is measured from 1 through 100 MHz (not 100 Mbps).
- **Category 6**
  The specifications for Category 6 aim to deliver a 100 m (330 feet) channel of twisted pair cabling that provides a minimum ACR at 200-250 MHz that is approximately equal to the minimum ACR of a Category 5 channel at 100 MHz.
  Category 6 includes all of the CAT 5e parameters but sweeps the test frequency out to 200 MHz, greatly exceeding current Category 5 requirements The IEEE has proposed extending the test frequency to 250 MHz to characterize links that may be marginal at 200 MHz.
  Test parameters: All of the same performance parameters that have been specified for Category 5e.
  Frequency range for specifications: Category 6 components and links are to be tested to 250 MHz even though the ACR values for the installed links are negative at 250 MHz.
  Note: Several vendors are promoting 'proprietary category 6 solutions'. These 'proprietary category 6' cabling systems only deliver a comparable level of performance if every component of connecting hardware (socket and plug combination) is purchased from the same vendor and from the same product series.

Specially selected 8-pin modular connector jacks and plugs need to be matched because they are designed as a 'tuned pair' in order to achieve the high level of cross-talk performance for NEXT and FEXT. If the user mixes and matches connector components, the system will no longer deliver the promised 'category 6-like' performance.

- **Category 7**
  This is a proposed shielded twisted pair (STP) standard that aims to support transmission up to 600 MHz.

11.7.6 Category 3, 4 and 5 performance features

Twisted pair cable is categorized in terms of its electrical performance properties. The features that characterize the data grades of UTP cable are defined in EIA/TIA 568:

Attenuation

This value indicates how much power the signal loses and is dependant on the frequency of the transmission. The maximum attenuation per 1000 feet of UTP cable at 20º Celsius at various frequencies is specified as follows:

Since connectors are needed at each end of the cable, the standards specify a worst-case attenuation figure to be met by the connecting hardware assuming they have characteristic impedance of 100 ohms to match the UTP cable.

Frequency Mhz	Cat. 3 dB	Cat. 4 dB	Cat.5 dB
0.772	**6.8**	**5.7**	**5.5**
1.0	**7.8**	**6.5**	**6.3**
4.0	**17.0**	**13**	**13**
8.0	**26**	**19**	**18**
10	**30**	**22**	**20**
16	**40**	**27**	**25**
20	**--**	**31**	**28**
25	**--**	**--**	**32**
31.25	**--**	**--**	**30**
62.5	**--**	**--**	**52**
100	**--**	**--**	**67**

Table 11.4
Maximum attenuation per 1000 feet for Cat 3, 4, 5 cables

Frequency Mhz	Cat. 3 dB	Cat. 4 dB	Cat.5 dB
1.0	0.4	0.1	0.1
4.0	0.4	0.1	0.1
8.0	0.4	0.1	0.1
10	0.4	0.1	0.1
16	0.4	0.2	0.2
20	--	0.2	0.2
25	--	--	0.2
31.25	--	--	0.2
62.5	--	--	0.3
100	--	--	0.4

Table 11.5
Worst case of connecting hardware for 100 ohm UTP

Mutual capacitance

Cable capacitance is measured in capacitance per unit length e.g. pF/ft, and lower values indicate better performance. The standards equate to mutual capacitance (measured at 1 kHz and 20 ^{0}C) for Category 3 cable not exceeding 20 pF/ft and for categories 4 and 11 not exceeding cables 17 pF/ft.

Characteristic impedance

All UTP cable should have characteristic impedance of 100 ±15 Ohms over the frequency range from 1 MHz to the cables highest frequency rating. Note these measurements need to be made on a cable of length at least one-eighth of a wavelength.

NEXT

The near end crosstalk (NEXT) indicates the degree of interference from a transmitting pair to an adjacent passive pair in the same cable at the near (transmission) end. This is measured by applying a balanced signal to one pair of wires and measuring its disturbing effect on another pair, both of which are terminated in their nominal characteristic impedance of 100 Ohms. This was shown in Figure 11.2 earlier in the chapter.

NEXT is expressed in decibels, in accordance with the following formula:

NEXT = 10 log P_d / P_x

Where

P_d power of the disturbing signal

P_x power of the cross talk signal

NEXT depends on the signal frequency and cable category. Performance is better at lower frequencies and for cables in the higher categories. Higher NEXT values indicate small crosstalk interference.

The standard specifies minimum values for NEXT for the fixed 10BaseT cables, known as horizontal UTP cable and for the connecting hardware. The following tables show these values for the differing categories of cable at various frequencies.

Since each cable has a connector at each end, the contribution to the NEXT from these connectors can be significant as shown in the following figure:

Frequency Mhz	Cat. 3 dB	Cat. 4 dB	Cat.5 dB
0.15	54	68	74
0.772	43	58	64
1.0	41	56	62
4.0	32	47	53
8.0	28	42	48
10	26	41	47
16	23	38	44
20	--	36	42
25	--	--	41
31.25	--	--	40
62.5	--	--	35
100	--	--	32

Table 11.6
Minimum NEXT for horizontal UTP cable at 20ºC

Frequency Mhz	Cat. 3 dB	Cat. 4 dB	Cat.5 dB
1.0	58	65	65
4.0	46	58	65
8.0	40	52	62
10	38	50	60
16	34	46	56
20	--	44	54
25	--	--	52
31.25	--	--	50
62.5	--	--	44
100	--	--	40

Table 11.7
Minimum NEXT for connectors at 20ºC

Note that the twists in the UTP cable, which enhance its cross talk performance, need to be removed to align the conductors in the connector. To maintain adequate NEXT performance the amount of untwisted wire and the separation between the conductor pairs should be minimized. The amount of untwisting should not exceed 13 mm (0.5 inch) for category 5 cables and 25 mm (1 inch) for category 4 cables.

Structural return loss (SRL)

The structural return loss (SRL) is a measure of the degree of mismatch between the characteristic impedance of the cable and the connector. This is measured as the ratio of the input power to the reflected signal power.

SRL = 10 log (input power/reflected power) dB

Higher values are better implying less reflection. For example 23 dB SRL corresponds to a reflected signal of seven percent of the input signal.

Frequency Range	Cat. 4	Cat. 5	% Reflected
1 to 20	23 dB	23 dB	7 %
20 to 100	--	14 dB	20 %

Table 11.8
Minimum structural return loss (SRL) at 20ºC

Direct current resistance

The DC resistance is an indicator of the ability of the connectors to transmit DC and low frequency currents. The maximum resistance between the input and output connectors, excluding the cable, is specified as 0.3 Ohm for Category 3, 4 and 5 UTP cables.

Ground plane effects

Note that if cables are installed on a conductive ground plane, such as a metal cable tray or in a metal conduit, the transmission line properties of mutual capacitance, characteristic impedance, return loss and attenuation can become two or three percent worse. This is not normally a problem in practice.

11.7.7 Advantages of twisted pair cable

Twisted pair cable has the following advantages over other types of cables for networks:

- It is easy to connect devices to twisted pair cable
- STP and ScTP do a reasonably good job of blocking interference
- UTP is quite inexpensive
- UTP is very easy to install
- UTP may already be installed (but make sure it all works properly and that it meets the performance specifications a network requires)

11.7.8 Disadvantages of twisted pair cable

Twisted pair cable has the following disadvantages:

- STP is bulky and difficult to work with
- UTP is more susceptible to noise and interference than coaxial or fiber optic cable
- UTP signals cannot go as far as they can with other cable types before they need amplification
- Skin effect can increase attenuation. This occurs when transmitting data at a fast rate over twisted pair wire. Under these conditions, the current tends to flow mostly on the outside surface of the wire. This greatly decreases the cross-section of the wire being used, and thereby increases resistance. This, in turn, increases signal attenuation

11.7.9 Selecting and installing twisted pair cable

When deciding on a category of cable, take future developments in the network and in technology into account. It is better to install Cat5e if Cat5 currently suited to the needs. Do not, however, install Cat6 if Cat5e is sufficient, as the bandwidth of Cat6 opens the door to unwanted interference, especially in industrial environments.

Check the wiring sequence before purchasing the cable. Different wiring sequences can hide behind the same modular plug in a twisted pair cable. (A wiring sequence, or wiring scheme, describes how wires are paired up and which locations each wire occupies in the plug.) If a plug that terminates one wiring scheme into a jack that continues with a different sequence is connected, the connection may not provide reliable transmission. If existing cable uses an incompatible wiring scheme, then a ‘cross wye’ as an adapter between the two schemes can be used.

If any of the cable purchases include patch cables (for example, to connect a computer to a wall plate), be aware that these cables come in straight through or reversed varieties. For networking applications, use the straight through cable, which means that wire 1 coming in connects to wire 1 going out. In a reversed cable; wire 2 connects to wire 2 rather than to wire 7, and so on.

11.8 Fiber optic cable

Fiber optic communication uses light signals and so transmissions are not subject to electromagnetic interference. Since a light signal encounters little resistance on its path (compared to an electrical signal traveling along a copper wire), this means that fiber optic cable can be used for much longer distances before the signal must be amplified, or

repeated. Some fiber optic segments can be several kilometers long before a repeater is needed.

In principle, data transmission using a fiber optic cable is many times faster than with copper and speeds of over 10 Gbps are possible. In reality, however, this advantage is nebulous because we are still waiting for the transmission and reception technology to catch up. Nevertheless, fiber optic connections deliver transmissions that are more reliable over greater distances, although at a somewhat greater cost. Cables of this type differ in their physical dimensions and composition and in the wavelength(s) of light with which the cable transmits.

Fiber optic cables are generally cheaper than coaxial cables, especially when comparing data capacity per unit cost. However, the transmission and receiving equipment, together with more complicated methods of terminating and joining these cables, makes fiber optic cable the most expensive medium for data communications.

The main benefits of fiber optic cables are:

- Enormous bandwidth (greater information carrying capacity)
- Low signal attenuation (greater speed and distance characteristics)
- Inherent signal security
- Low error rates
- Noise immunity (impervious to EMI and RFI)
- Logistical considerations (light in weight, smaller in size)
- Total galvanic isolation between ends (no conductive path)
- Safe for use in hazardous areas
- No crosstalk

11.8.1 Theory of operation

A fiber optic system has three components – light source, transmission medium, and detector. A pulse of light indicates one bit and absence of light indicates zero bits. The transmission medium is an ultra-thin fiber of glass. The detector generates an electrical impulse when light falls on it. A light source at one end of the optical fiber and a detector at the other end then results in a unidirectional data transmission system that accepts an electrical signal, converts it into light pulses, and then re-converts the light output to an electrical signal at receiving end.

A property of light, called refraction (bending of a ray of light when it passes from one medium into other medium) is used to ‘guide’ the ray through the whole length of fiber. The amount of refraction depends on media, and above a certain critical angle, all rays of light are refracted back into the fiber, and can propagate in this way for many kilometers with little loss.

There will be many rays bouncing around inside the optical conductor, each ray is said to have a different mode. Such a conductor is called multi-mode fiber. If the conductor diameter is reduced to within a few wavelengths of light, then the light will propagate only in a single ray. Such a fiber is called single-mode fiber.

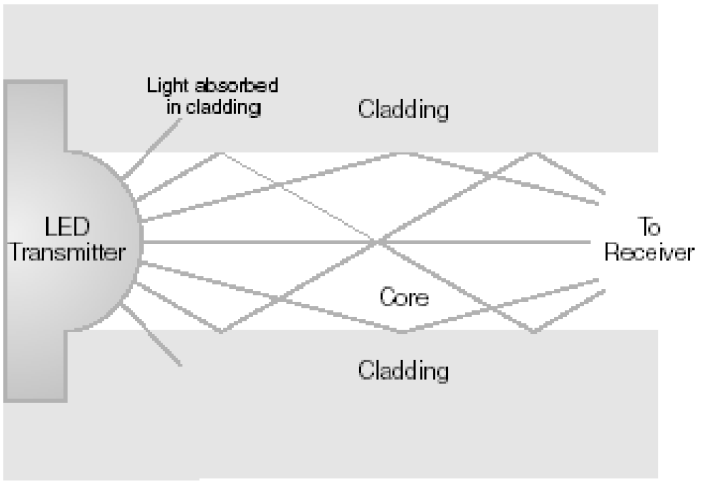


Figure 11.7
LED source coupled to a multi-mode fiber

11.8.2 Multimode fibers

The light takes many paths between the two ends as it reflects from the sides of the fiber core. This causes the light rays to arrive both out of phase and at different times resulting in a spreading of the original pulse shape. As a result, the original sharp pulses sent from one end become distorted by the time they reach the receiving end.

The problem becomes worse as data rates increase. Multimode fibers, therefore, have a limited maximum data rate (bandwidth) as the receiver can only differentiate between the pulsed signals at a low data rate. The effect is known as 'modal dispersion' and its result referred to as 'inter-symbol interference'. For slower data rates over short distances, multimode fibers are quite adequate and speeds of up to 300 Mbps are readily available.

A further consideration with multimode fibers is the 'index' of the fiber (how the impurities are applied in the core). The cable can be either 'graded index' (more expensive but better performance) or 'step index' (less expensive). The type of index affects the way in which the light waves reflect or refract off the walls of the fiber. Graded index cores focus the modes as they arrive at the receiver, and consequently improve the permissible data rate of the fiber.

The core diameters of multimode fibers typically range between 50–100μm. The two most common core diameters are 50 and 62.5μm.

Multimode fibers are easier and cheaper to manufacture than mono-mode fibers. Multimode cores are typically 50 times greater than the wavelength of the light signal they will propagate. With this type of fiber, an LED transmitter light source is normally used because it can be coupled with less precision than a laser diode. With the wide aperture and LED transmitter, the multimode fiber will send light in multiple paths (modes) toward the receiver.

One measure of signal distortion is modal dispersion, which is represented in nanoseconds of signal spread per kilometer (ns/km). This value represents the difference in arrival time between the fastest and slowest of the alternate light paths. The value also imposes an upper limit on the bandwidth. With step-index fiber, expect between 15 and 30 ns/km. Note that a modal dispersion of 20 ns/km yields a bandwidth of less than 50 Mbps.

11.8.3 Monomode/single mode fibers

'Monomode' or 'single mode' fibers are less expensive to manufacture but more difficult to interface. They allow only a single path or mode for the light to travel down the fiber with minimal reflections. Monomode fibers typically use lasers as light sources.

Monomode fibers do not suffer from major dispersion or overlap problems and permit a very high rate of data transfer over much longer distances.

The core of the fibers is much thinner than multimode fibers at approximately 8.5µm. The cladding diameter is 125µm, the same as for multimode fibers.

Optical sources must be powerful and aimed precisely into the fiber to overcome any misalignment (hence the use of laser diodes). The thin monomode fibers are difficult to work with when splicing, terminating, and are consequently expensive to install.

Single-mode fiber has the least signal attenuation, usually less than 0.25dB per kilometer. Transmission speeds of 50 Gbps and higher are possible.

Even though the core of single-mode cable is shrunk to very small sizes, the cladding is not reduced accordingly. For single-mode fiber, the cladding is made the same size as for the popular multimode fiber optic cable. This both helps create a *de facto* size standard and also makes the fiber and cable easier to handle and more resistant to damage.

Specification of optical fiber cables

Optical fibers are specified based on diameter. A fiber specified as 50/150 has a core of 50 µm and a cladding diameter of 150 µm. The most popular sizes of multimode fibers are 50/125, used mainly in Europe, and 62.5/125, used mainly in Australia and the USA.

Another outer layer provides an external protection against abrasion and shock. Outer coatings can range from 250 - 900 µm in diameter, and very often cable specifications include this diameter, for example: 50/150/250.

To provide additional mechanical protection, the fiber is often placed inside a loose, but stiffer, outer jacket which adds thickness and weight to the cable. Cables made with several fibers are most commonly used. The final sheath and protective coating on the outside of the cable depends on the application and where the cable will be used. A strengthening member is normally placed down the center of the cable to give it longitudinal strength. This allows the cable to be pulled through a conduit or hung between poles without causing damage to the fibers. The tensile members are made from steel or Kevlar, the latter being more common. In industrial and mining applications, fiber cores are often placed inside cables used for other purposes, such as trailing power cables for large mining, stacking or reclaiming equipment.

Experience has shown that optic fibers are likely to break during a 25-year provision period. In general, the incremental cost of extra fiber cores in cables is not high when compared to overall costs (including installation and termination costs). So it is usually worthwhile specifying extra cores as spares, for future use.

Joining optical fibers

In the early days of optic fibers, connections and terminations were a major problem. Largely, this has improved but connections still require a great deal of care to avoid signal losses that will affect the overall performance of the communications system.

There are three main methods of splicing optic fibers:

- **Mechanical:** Where the fibers are fitted into mechanical alignment structures
- **Chemical:** Where the two fibers are fitted into a barrel arrangement with epoxy glue in it. They are then heated in an oven to set the glue
- **Fusion splicing:** Where the two fibers are heat-welded together

To overcome the difficulties of termination, fiber optic cables can be provided by a supplier in standard lengths such as 10 m, 100 m or 1000 m with the ends cut and finished with a mechanical termination ferrule that allows the end of the cable to slip into a closely matching female socket. This enables the optical fiber to be connected and disconnected as required.

The mechanical design of the connector forces the fiber into a very accurate alignment with the socket and results in a relatively low loss. Similar connectors can be used for in-line splicing using a double-sided female connector. Although the loss through this type of connector can be an order of magnitude greater than the loss of a fused splice, it is much quicker and requires less special tools and training. Unfortunately, mechanical damage or an unplanned break in a fiber requires special tools and training to repair and re-splice.

One way around this problem is to keep spare standard lengths of pre-terminated fibers that can quickly and easily be plugged into the damaged section. The techniques for terminating fiber optic cables are constantly being improved to simplify these activities.

Limitations of fiber optic cables

On the negative side, the limitations of fiber optic cables are as follows:

- Cost of source and receiving equipment is relatively high
- It is difficult to 'switch' or 'tee-off' a fiber optic cable so fiber optic systems are most suitable for point-to-point communication links
- Techniques for joining or terminating fibers (mechanical and chemical) are difficult and require precise physical alignment. Special equipment and specialized training are required
- Equipment for testing fiber optic cables is different and more expensive than traditional methods used for electronic signals
- Fiber optic systems are used almost exclusively for binary digital signals and are not really suitable for long distance analog signals

Uses of optical fiber cables

Optical fiber cables are used less often to create a network than to connect together two networks or network segments. For example, cable that must run between floors is often fiber optic cable, most commonly of the 62.5/125 varieties with an LED (light-emitting diode) as the light source.

Being impervious to electromagnetic interference, fiber is ideal for such uses because the cable is often run through the lift, or elevator, shaft, and the drive motor puts out strong interference when the cage is running.

A disadvantage of fiber optic networks has been price. Network interface cards (NICs) for fiber optic nodes can cost many times the cost of some Ethernet and ARCnet cards. It is not always necessary, however, to use the most expensive fiber optic connections. For short distances and smaller bandwidth, inexpensive cable is adequate. Generally, a fiber optic cable will always allow a longer transmission than a copper cable segment.

11.8.4 Fiber optic cable components

The major components of a fiber optic cable are the core, cladding, buffer, strength members, and jacket. Some types of fiber optic cable even include a conductive copper wire that can be used to provide power for example, to a repeater.

Fiber optic core and cladding

The core of fiber optic cable consists of one or more glass or plastic fibers through which the light signal moves. Plastic is easier to manufacture and use but works over shorter distances than glass.

In networking contexts, the most popular core sizes are 50, 62.5 and 100 microns. Most of the fiber optic cable used in networking has two core fibers: one for communicating in each direction.

The core and cladding are actually manufactured as a single unit. The cladding is a protective layer of glass or plastic with a lower index of refraction than the core. The lower index means that light that hits the core walls will be redirected back to continue on its path. The cladding will be anywhere between a hundred microns and a millimeter (1000 microns) or so.

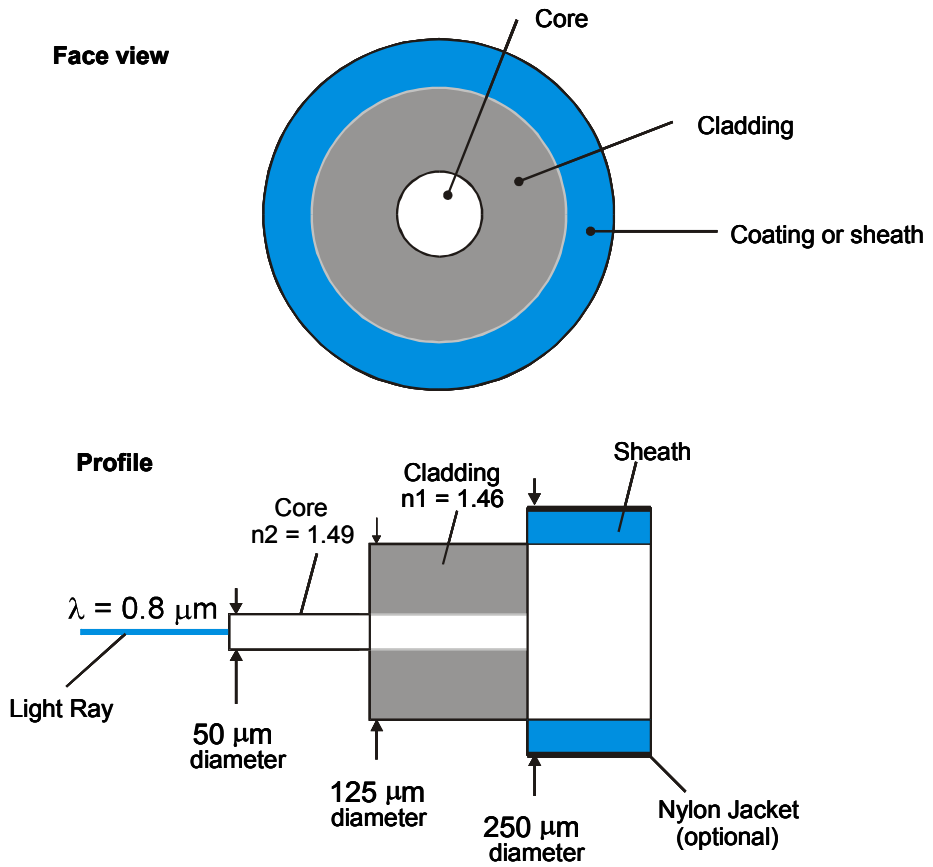


Figure 11.8
Fiber optic cable components

Fiber optic buffer

The buffer of a fiber optic cable is a one or more layer of plastic surrounding the cladding. The buffer helps strengthen the cable, thereby decreasing the likelihood of micro cracks, which can eventually grow into larger breaks in the cable. The buffer also protects the core and cladding from potential corrosion by water or other materials in the operating environment. The buffers can double the diameter of some cable.

A buffer can be loose or tight. A loose buffer is a rigid tube of plastic with one or more fibers (consisting of core and cladding) running loosely through it. The tube takes on all the stresses applied to the cable, buffering the fiber from these stresses. A tight buffer fits

snugly around the fiber(s). A tight buffer can protect the fibers from stress due to pressure and impact, but not from changes in temperature. Loose-buffered cables are normally used for external applications while tight-buffered fibers are usually restricted to internal cables.

Strength members

Fiber optic cable also has strength members, which are strands of very tough material (such as steel, fiberglass or Kevlar) that provide extra strength for the cable. Each of the substances has advantages and drawbacks. For example, steel attracts lightning, which will not disrupt an optical signal but may seriously damage the equipment or the operator sending or receiving such a signal.

Fiber optic jacket

The jacket of a fiber optic cable is an outer casing that can be plenum or non-plenum, as with electrical cable. In cable used for networking, the jacket usually houses at least two fiber/cladding pairs: one for each direction.

11.8.5 Fiber core refractive index changes

One reason why optical fiber makes such a good transmission medium is that the different indexes of refraction for the cladding and core help to contain the light signal within the core, producing a wave-guide for the light. Cable can be constructed by changing abruptly or gradually from the core refractive index to that of the cladding. The two major types of multimode fiber differ in this feature.

Step-index cable

Cable with an abrupt change in refraction index is called step-index cable. In step-index cable, the change is made in a single step. Single-step multimode cable uses this method, and it is the simplest, least expensive type of fiber optic cable. It is also the easiest to install. The core is usually between 50 and 62.5 microns in diameter; the cladding is at least 125 microns. The core width gives light quite a bit of room to bounce around in, and the attenuation is high (at least for fiber optic cable): between 10 and 50 dB/km. Transmission speeds between 200 Mbps and 3 Gbps are possible, but actual speeds are much lower.

Graded-index cable

Cable with a gradual change in refraction index is called graded-index cable, or graded-index multimode. This fiber optic cable type has a relatively wide core, like single-step multimode cable. The change occurs gradually and involves several layers, each with a slightly lower index of refraction. A gradation of refraction indexes controls the light signal better than the step-index method. As a result, the attenuation is lower, usually less than 5 dB/km. Similarly, the modal dispersion can be 1 ns/km and lower, which allows more than ten times the bandwidth of step-index cable. Graded-index multimode cable is the most commonly used type for network wiring.

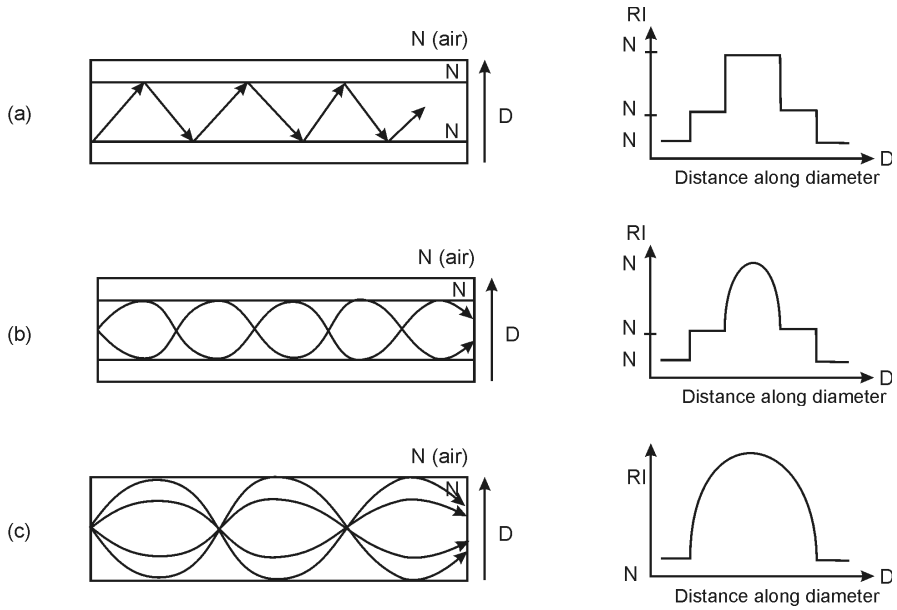


Figure 11.9
Fiber refractive index profiles

Fiber composition

Fiber core and cladding may be made of plastic or glass. The following list summarizes the composition combinations, going from highest quality to lowest:

- **Single-mode glass**: has a narrow core, so only one signal can travel through
- **Graded-index glass**: this is a multi-mode fiber, and the gradual change in refractive index helps give more control over the light signal and significantly reduces modal dispersion
- **Step-index glass**: this is also multi-mode. The abrupt change from the refractive index of the core to that of the cladding means the signal is less controllable, producing low bandwidth fibers
- **Plastic-coated silica (PCS):** has a relatively wide core (200 microns) and a relatively low bandwidth (20 MHz)
- **Plastic:** this should be used only for low speed (e.g. 56k bps) over short distances (15 m)

To summarize, fiber optic cables may consist of a glass core and glass cladding (the best available). Glass yields much higher performance, in the form of higher bandwidth over greater distances. Single-mode glass with a small core is the highest quality. Cables may also consist of glass core and plastic cladding. Finally, the lowest grade fiber composition is plastic core and plastic cladding. Step-index plastic has the lowest performance.

Fiber optic cable quality

Some points about fiber optic cable quality:

- The smaller the core, the better the signal
- Fiber made of glass is better than fiber made of plastic
- The purer and cleaner the light, the better the signal. (Pure, clean light is a single color, with minimal spread around the primary wavelength of the color)
- Certain wavelengths of light behave better than others

Fiber optic cable designations

Fiber optic cables are specified in terms of their core and cladding diameters. For example, a 62.5/125 cable has a core with a 62.5-micron diameter and cladding with twice that diameter.

Following are some commonly used fiber optic cable configurations:

- **8/125:** A single-mode cable with an 8-micron core and a 125-micron cladding. Systems using this type of cable are expensive and currently used only in contexts where extremely large bandwidths are needed (such as in some real-time applications) and/or where large distances are involved. An 8/125-cable configuration is likely to use a light wavelength of 1300 or 1550 nm
- **62.5/125:** The most popular fiber optic cable configuration, used in most network applications. Both 850 and 1300 nm wavelengths can be used with this type of cable
- **100/140:** The configuration that IBM first specified for fiber optic wiring for a Token Ring network. Because of the tremendous popularity of the 62.5/125 configurations, IBM now supports both configurations

If purchasing fiber optic cable, it is important that the correct core size is bought. If the type of desired network is determined, constrains of a particular core size will arise.

Advantages of fiber optic cables

Fiber optic connections offer the following advantages over other types of cabling systems:

- Light signals are impervious to interference from EMI or electrical cross talk.
- Light signals do not interfere with other signals. As a result, fiber optic connections can be used in extremely adverse environments, such as in lift shafts or assembly plants, where powerful motors and engines produce lots of electrical noise.
- Fiber optic lines are much harder to tap into, so they are more secure for private lines.
- Light has a much higher bandwidth, or maximum data transfer rate, than electrical connections. This speed advantage is not always achieved in practice, however.
- The signal has a much lower loss rate, so it can be transmitted much further than it could be with coaxial or twisted pair cable before amplification is necessary.
- Optical fiber is much safer, because there is no electricity and so no danger of electrical shock or other electrical accidents. However, if a laser source is used, there is danger of eye damage.
- Fiber optic cable is generally much thinner and lighter than electrical cable, and so it can be installed more unobtrusively. (Fiber optic cable weighs about 30 grams per meter; coaxial cable weighs nearly ten times that much).
- The installation and connection of the cables is nowadays much easier than it was at first.

Disadvantages of fiber optic cable

The disadvantages of fiber optic connections include the following:

- Fiber optic cable is more expensive than other types of cable.

- Other components, particularly ‘fiber’ NICs are expensive.
- Certain components, particularly couplers, are subject to optical cross talk.
- Fiber connectors are not designed to use incessantly. Generally, they are designed for fewer than a thousand mating. After that, the connection may become loose, unstable, or misaligned. The resulting signal loss may be unacceptably high.
- Many more parts can break in a fiber optic connection than in an electrical one.

11.9 The IBM cable system

IBM designed the IBM cable system for use in its token ring networks and also for general-purpose premises wiring. IBM has specified nine types of cable, mainly twisted pair, but with more stringent specifications than for the generic twisted pair cabling. The types also include fiber optic cable, but exclude coaxial cable.

The twisted pair versions differ in the following ways:

- Whether the type is shielded or unshielded
- Whether the carrier wire is solid or stranded
- The gauge (diameter) of the carrier wire
- The number of twisted pairs

11.9.1 IBM type 1 cable specifications

Specifications have been created for seven of the nine types with types 4 and 7 undefined. However, the only type relevant to Ethernet users is type 1, as it may be used instead of the usual EIA/TIA-type UTP cable, provided the appropriate impedance-matching baluns are employed.

Type 1 cable is shielded twisted pair, with two pairs of 22-gauge solid wire. Its impedance is 150 ohms and the maximum frequency allowed is 16 MHz. Although not required by the specifications, a plenum version is also available, at about twice the cost of the non-plenum cable. IBM specification numbers are 4716748 for non-plenum data cable, 4716749 for plenum data cable, 4716734 for outdoor data cable, and 6339585 for riser cable.

Type 1A cable

This consists of two ‘data grade’ shielded twisted pairs and uses 22 AWG solid conductors. Its impedance is 150 ohms and the maximum allowed frequency is 300 MHz. IBM specification numbers are 33G2772 for non-plenum data cable, 33G8220 for plenum data cable, and 33G2774 for riser cable.

11.10 Ethernet cabling requirement overview

This section now lists in brief cabling requirements and minimum specification followed in industry.

10BaseT

Type of cable	Cat. 3,4 or 5 UTP
Maximum length	100 m
Max. impedance allowed	75–165 ohms
Max. attenuation allowed	11.5 db at 5–10 MHz
Max. jitter allowed	5 ns
Delay	1 microsecond
Crosstalk	60 dB for a 10 MHz link
Other considerations	Use plenum cable for ambients higher than 20ºC

Table 11.9
10BaseT requirements

10Base2

Type of cable	Thin coaxial
Maximum length	185 m
Max. number of stations	30
Max. impedance allowed	50 ohms
Other considerations	Termination at both ends required using 50 ohm terminators

Table 11.10
10Base 2 Cable requirements

10Base5

Type of cable	Thick coaxial
Maximum length	500 m
Max. number of stations	100
Max. impedance allowed	50 ohms
Other considerations	Termination at both ends required using 50 ohm terminators

Table 11.11
10Base5 cable requirements

10BaseT full-duplex

Type of cable	Cat. 3,4 or 5 UTP
Maximum length	100 m
Max. impedance allowed	75–165 ohms
Max. attenuation allowed	11.5 db at 5–10 MHz
Max. jitter allowed	5 ns
Delay	Delay is not a factor
Crosstalk	60 dB for a 10 MHz link
Other considerations	Use plenum cable for ambients higher than 20ºC

Table 11.12
10BaseT Full duplex cabling requirements

10BaseF multimode

Type of cable	50/125 or 62.5/125 or 100/140 micron multimode fiber
Maximum length	2 km
Max. attenuation allowed	<13 dB for 50/125 micron <16 dB for 62.5/125 micron <19 dB for 100/140 micron
Max. delay	Total 25.6 microseconds one way

Table 11.13
10BaseF Multi-mode cabling requirements

100BaseFX multimode cabling requirements

Type of cable	50/125 or 62.5/125 or 100/140 micron multimode fiber
Maximum length	2 km for simplex, 412 m for duplex
Max. attenuation allowed @ 850 nm	<13 dB for 50/125 micron <16 dB for 62.5/125 micron <19 dB for 100/140 micron
Max. delay	Total 25.6 microseconds one way

Table 11.14
100BaseFX Cabling requirements

100Base TX

Type of cable	Cat. 5 UTP cable
Maximum length	100 m
Max. attenuation allowed @ 100 MHz	240 dB
Max. delay	Total 1 microseconds
Max. impedance allowed	75–165 ohms
Max. allowed crosstalk	27 dB

Table 11.15
00BaseTX cabling requirements

11.11 Cable connectors

11.11.1 AUI cable connectors

AUI cable is uses DB15 connectors.

The DB15 connector (male or female) provides 15 pins or channels depending in gender. The mapping of these 15 pins of the AUI cable has been dealt with in section 4.1.

11.11.2 Coaxial cable connectors

A segment of coaxial cable has a connector at each end. The cable is attached through these end connectors to a T connector, a barrel connector, another end connector, or to a terminator. Through these connectors, another cable or a hardware device is attached to the coaxial cable. In addition to their function, connectors differ in their attachment mechanism and components. For example, BNC connectors join two components by

plugging them together and then turning the components to click the connection into place. Different size coaxial cables require different sized connectors, matched to the characteristic impedance of the cable, so the introduction of connectors causes minimal reflection of the transmitted signals.

For coaxial cable, the following types of connectors are available:

- N series connectors are used for thick coaxial cable
- BNC is used for thin coaxial cable
- TNC (threaded nut connector) may be used in the same situations as a BNC, provided that the other connector is also using TNC

N Type

These connectors are used for the termination of thick coaxial cables and for the connection of transceivers to the cable. When used to provide a transceiver tap, the coaxial cable is broken at an annular ring and two N type connectors are attached to the resulting bare ends. These N type connectors, once in place, are screwed onto a barrel housing.

The barrel housing contains a center channel across which the signals are passed and a pin or cable that contacts this center channel, providing access to and from the core of the coaxial cable. The pin that contacts the center channel is connected to the transceiver assembly and provides the path for signal transmission and reception.

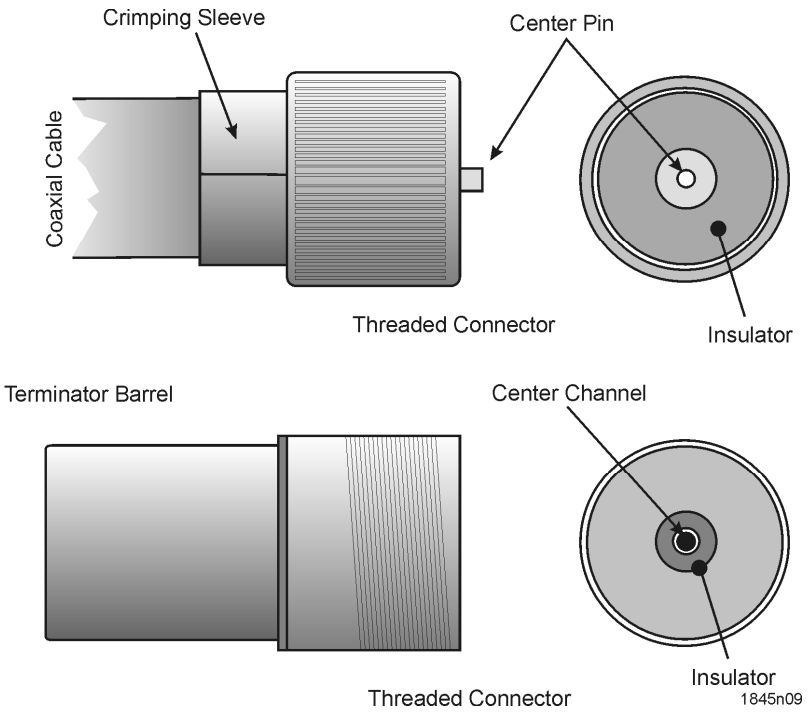


Figure 11.10
N Type connector and terminator

Thick coaxial cables require termination with N type connectors. As the coaxial cable carries network transmissions as voltage, both ends of the thick coaxial cable must be terminated to keep the signal from reflecting throughout the cable, which would disrupt network operation. The terminators used for thick coaxial cable are 50-Ohm. These terminators are screwed into an N type connector placed at the end of a run.

BNC

The BNC connector, used for 10Base2, is an intrusive connector much like the N type connector used with thick coaxial cable. The BNC connector (shown in Figure 11.11) requires that the coaxial cable be broken to make the connection. Two BNC connectors are either screwed onto or crimped to the resulting bare ends.

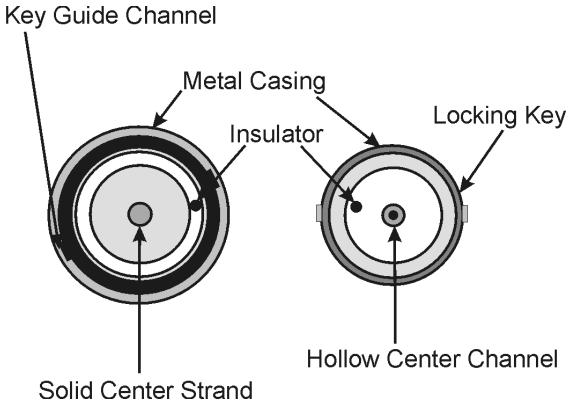


Figure 11.11
BNC connector

BNC male connectors are attached to BNC female terminators or T connectors (Figure 11.12). The outside metal housing of the BNC male connector has two guide channels that slip over corresponding locking key posts on the female BNC connector. When the outer housing is placed over the T connector and turned, the connectors will snap securely into place.

Tapping of coax cable

Tapping a thick coaxial cable is done without breaking the cable itself by use of a non-intrusive, or 'vampire' tap (Figure 11.12). This tap inserts a solid pin through the thick insulating material and shielding of the coaxial cable. The solid pin reaches in through the insulator to the core wire where signals pass through the cable. The signals travel through the pin to and from the core.

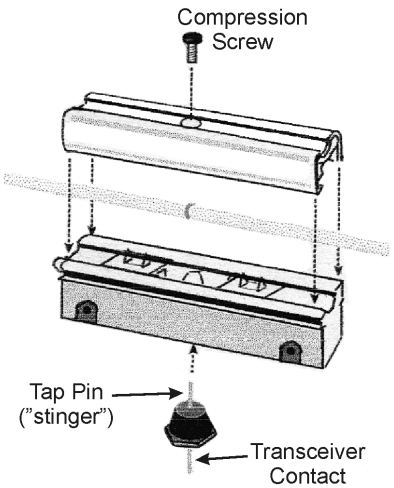


Figure 11.12
'Vampire' non-intrusive tap and cable saddle

Non-intrusive taps are made up of saddles, which bind the connector assembly to the cable, and tap pins, which burrow through the insulator to the core wire.

Non-intrusive connector saddles are clamped to the cable to hold the assembly in place, and usually are either part of, or are easily connected to, an Ethernet transceiver assembly. (See figure 11.13) The non-intrusive tap's cable saddle is then inserted into a transceiver assembly. The contact pin, that carries the signal from the tap pin's connection to the coaxial cable core, makes a contact with a channel in the transceiver housing. The transceiver breaks the signal up and carries it to a DB15 connector, to which an AUI cable may be connected.

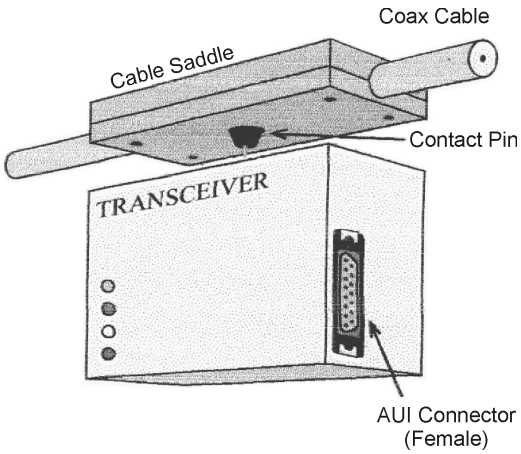


Figure 11.13
Cable saddle and transceiver assembly

Thin coax T connectors

Connections from the cable to network nodes are made using T connectors, which provide taps for additional runs of coaxial cable to workstations or network devices. T connectors, as shown in figure 11.14 below, provide three BNC connections, two of which attach to male BNC connectors on the cable itself and one of which is used for connection to the female BNC connection of a transceiver or network interface card on a workstation.

T connectors should be attached directly to the BNC connectors of network interface cards or other Ethernet devices.

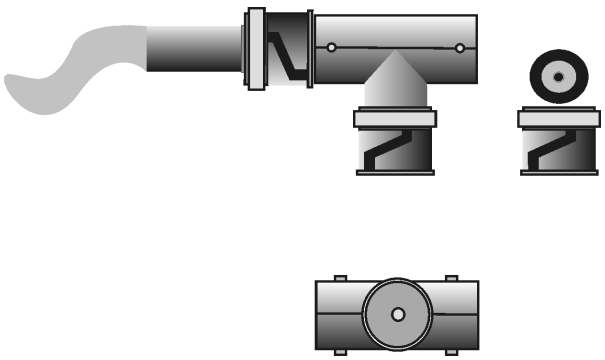


Figure 11.14
Thin coax T connector

The use of the crimp-on BNC connectors is recommended for more stable and consistent connections. BNC connectors use the same pin-and-channel system to provide a contact that is used in the thick coaxial N type connector.

The so-called crimpless connectors should be avoided at all costs. A good quality-crimping tool is very important for BNC connectors. The handle and jaws of the tool should have a ratchet action, to ensure the crimp is made to the required compression. Ensure that the crimping tool has jaws wide enough to cover the entire crimped sleeve at one time. Anything less is asking for problems.

The typical crimping sequence is normally indicated on the packaging with the connector. Ensure that the center contact is crimped onto the conductor before inserting into the body of the connector.

Typical dimensions are shown in the Figure 11.15 below.

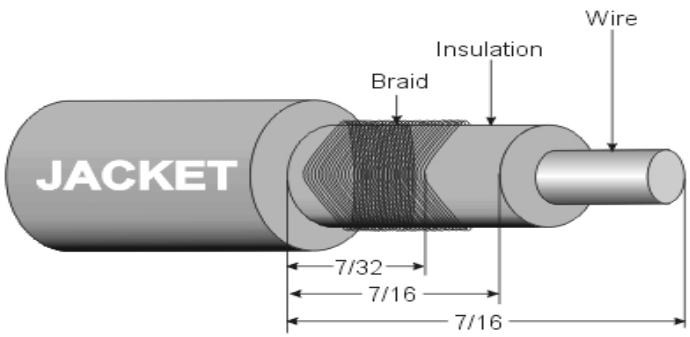


Figure 11.15
BNC coaxial cable termination

11.11.3 UTP cable connectors

RJ-45 cable connectors

The RJ-45 connector is a modular, plastic connector that is often used in UTP cable installations. The RJ-45 is a keyed connector, designed to plug into an RJ-45 port only in the correct alignment. The connector is a plastic housing that is crimped onto a length of UTP cable using a custom RJ-45 die tool. The connector housing is often transparent, and consists of a main body, the contact blades or pins, the raised key, and a locking clip and arm.

The 8-wire RJ-45 connector is small, inexpensive and popular. As a matter of interest, the RJ-45 is different to the RJ-11, although they look the same. RJ-45 is an eight-position plug or jack as opposed to the RJ-11, which is a six-position jack. ‘RJ’ stands for registered jack and is supposed to refer to a specific wiring sequence.

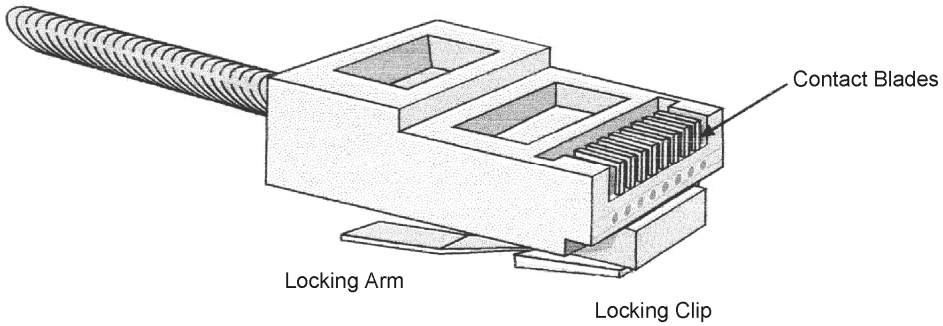


Figure 11.16
RJ-45 connector

The locking clip, part of the raised key assembly, secures the connector in place after a connection is made. When the RJ-45 connector is inserted into a port, the locking clip is pressed down and snaps up into place. A thin arm, attached to the locking clip, allows the clip to be lowered to release the connector from the port.

Stranded or solid conductors

RJ-45 connectors for UTP cabling are available in two basic configurations; stranded and solid. These names refer to the type of UTP cabling that they are designed to connect to. The blades of the RJ-45 connector end in a series of points that pierce the jacket of the wires and make the connection to the core. Different types of connections are required for each type of core composition.

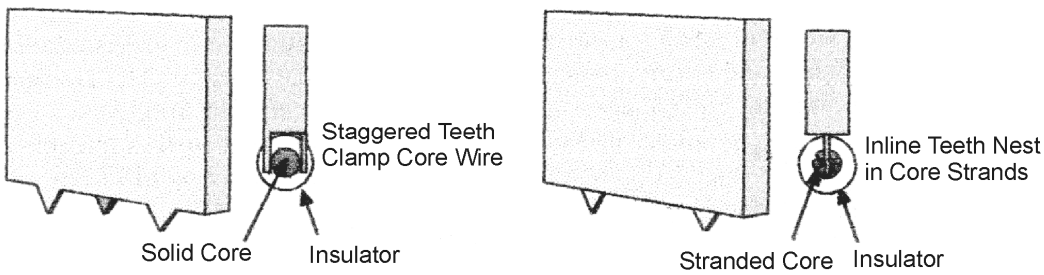


Figure 11.17
Solid and stranded RJ-45 blades

A UTP cable that uses stranded core wires will allow the contact points to nest among the individual strands. The contact blades in a stranded RJ-45 connector, therefore, are laid out with their contact points in a straight line. The contact points cut through the insulating material of the jacket and make contact with several strands of the core.

The solid UTP connector arranges the contact points of the blades in a staggered fashion. The purpose of this arrangement is to pierce the insulator on either side of the core wire and make contacts on either side. As the contact points cannot burrow into the solid core, they clamp the wire in the middle of the blade, providing three opportunities for a viable connection.

There are two terms often used with connectors:

- **Polarization** means the physical form and configuration of connectors
- **Sequence** refers to the order of the wire pairs

An RJ-45 crimping tool shown in Figure 11.18 is often referred to as a 'plug presser' because of its pressing action. When a cable is connected to the cable, the plastic connector is placed in a die in the jaw of the crimping tool. The wires are carefully dressed and inserted into the open connector and then the handles of the crimping tool are closed to force the connector together. The following section discusses the various pin-outs required for the RJ-45 connector.

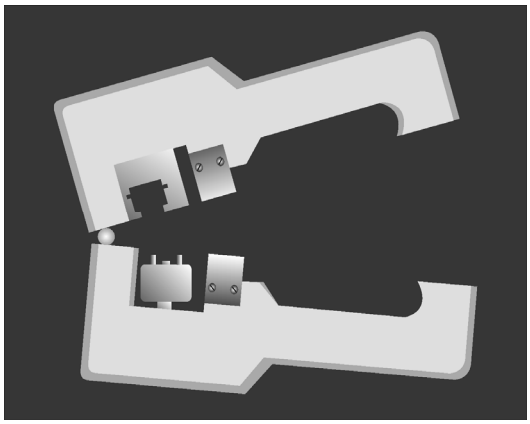


Figure 11.18
Crimping tool

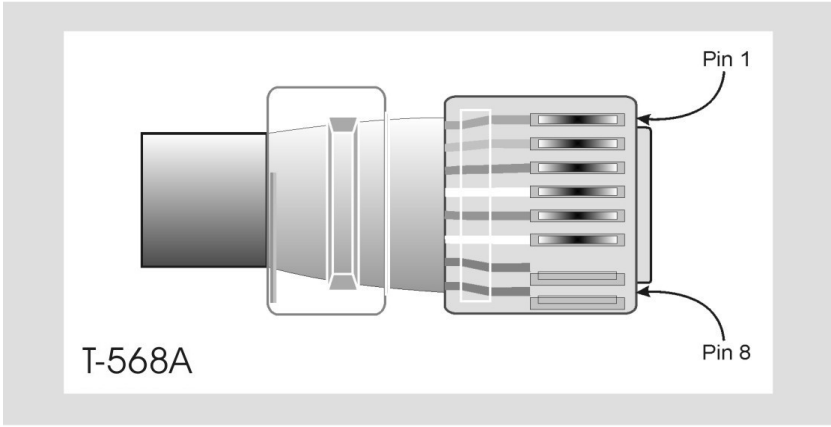
With UTP horizontal wiring, there is general agreement on how to color code each of the wires in a cable. There is however, no general agreement on the physical connections for mating UTP wires and connectors. There are various connector configurations in use, but the main ones are EIA/TIA 568A and EIA/TIA 568B. The difference between them is that the green and orange pairs are interchanged.

RJ-45 pin assignments

Table 11.17 shows pin assignments for each of the Ethernet twisted pair cabling systems:

Contact	10BaseT signal	100BaseTX signal	100BaseT4 signal	100BaseT2 signal	1000BaseT signal
1	TX+	TX+	TX D1+	B1 DA+	B1 DA+
2	TX–	TX–	TX D1–	B1 DA–	B1 DA–
3	RX+	RX+	RX D2+	B1 DB+	B1 DB+
4	Not used	Not used	B1 D3+	Not used	B1 DC+
5	Not used	Not used	B1 D3-	Not used	B1 DC–
6	RX–	RX–	RX D2-	B1 DB–	B1 DB–
7	Not used	Not used	B1 D4+	Not used	B1 DD+
8	Not used	Not used	B1 D4–	Not used	B1 DD–
TX= Transmit data B= Bidirectional data RX= Receive data					

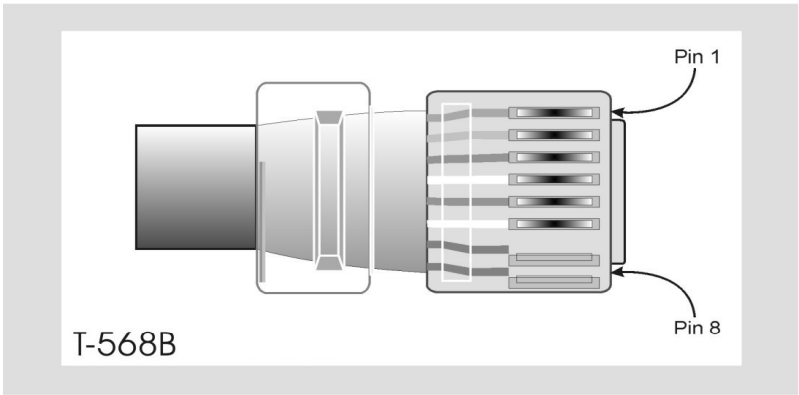
Table 11.17
RJ-45 pin assignments for various Ethernet twisted pair cables



Green/White	1
Green	2
Orange/White	3
Blue	4
Blue/White	5
Orange	6
Brown/White	7
Brown	8

Figure 11.19
T-568A pin assignments

The sequence is green-orange-blue-brown. Wires with white are always on the odd-numbered pins, and pins 3/6 "straddle" pins 4/5.



Orange/White	1
Orange	2
Green/White	3
Blue	4
Blue/White	5
Green	6
Brown/White	7
Brown	8

Figure 11.20
T-568 pin assignments

For T-568B, the orange and green pairs are swapped, so the sequence is now: orange-green-blue-brown, still with the white wires on odd-numbered pins.

If a crossover cable is required, the transmit and receive pairs (1/2 and 3/6) on the one end of the cable have too be swapped. The cable will therefore look like T-568A on one end and T-568B on the other end.

Medium independent interface (MII) connector

MII interface were discussed in Chapter 4 (see Figure 4.2) along with its pin assignments and pin functions.

RJ-21 connector for 25-pair UTP cable

25-pair UTP cable was briefly discussed earlier in this chapter. This cable is connected by RJ-21 connectors. The RJ-21 connector, also known as a 'Telco connector', is a D-shaped metal or plastic housing that is wired and crimped to a UTP cable made up of 50 wires, a 25-pair cable. The RJ-21 connector can only be plugged into an RJ-21 port. The connector itself is large, and the cables that it connects to are often quite heavy, so the RJ-21 relies on a tight fit and good cable management practices to keep itself in the port. Some devices may also incorporate a securing strap that wraps over the back of the connector and holds it tight to the port.

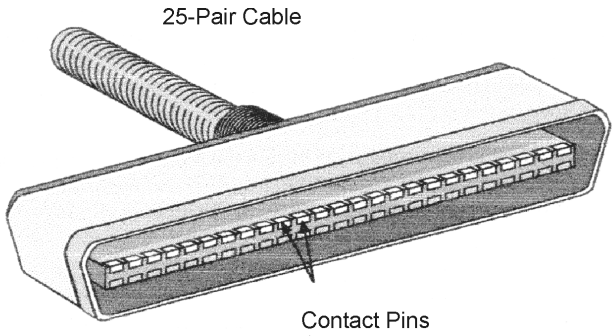


Figure 11.21
RJ-21 connector for 25-pair UTP cables

The RJ-21 is used in locations where 25-pair cable is being run either to stations or to an intermediary cable management device such as a patch panel or punch down block. Due to the bulk of the 25-pair cable and the desirability of keeping the wires within the insulating jacket, as much as possible, 25-pair cable is rarely run directly to Ethernet stations.

The RJ-21 connector, when used in a 10BaseT environment, must use the EIA/TIA 568A pin out scheme. The numbers of the RJ-21 connector's pins are detailed in Figure 11.24 below.

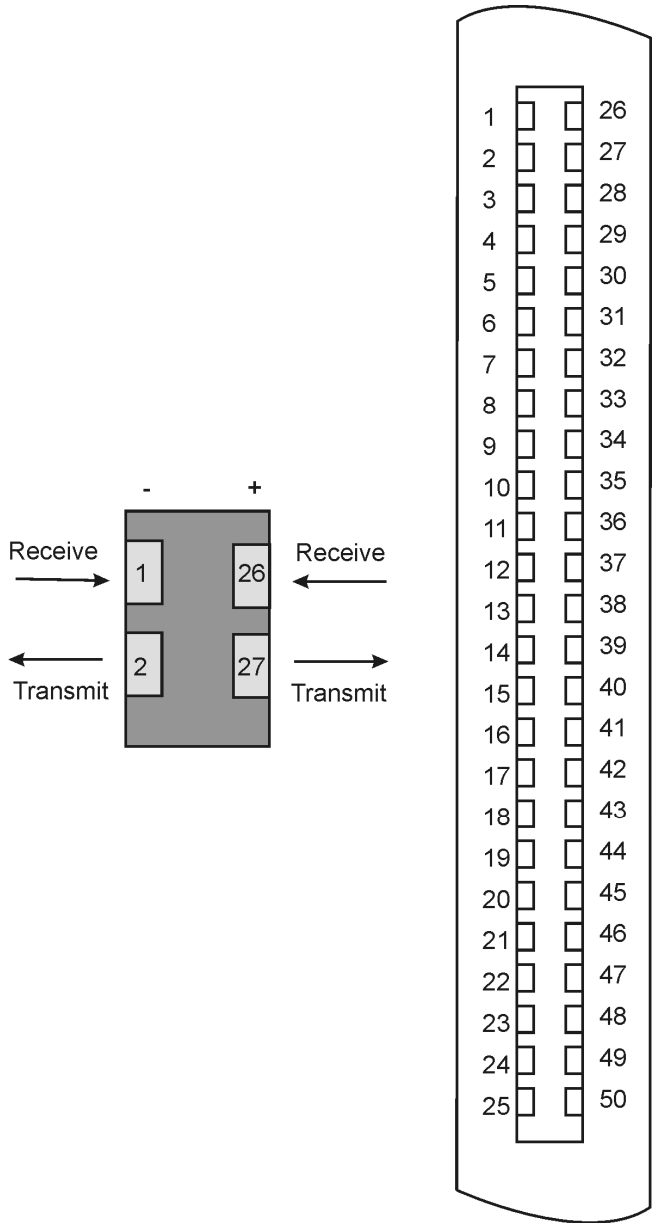


Figure 11.22
RJ-21 pin mapping for 10BaseT

Punch down blocks

While not strictly a connector type, the punch down block is a fairly common component in many Ethernet 10BaseT installations that use 25-pair cable. The punch downs are bayonet pins to which UTP wire strands are connected. The bayonet pins are arranged in 50 rows of four columns each. The pins that make up the punch down block are identified by the row and column they are members of.

Each of the four columns is lettered A, B, C, or D, from leftmost to rightmost. The rows are numbered from top to bottom, one to 50. Thus, the upper left hand pin is identified as A1, while the lower right hand pin is identified as D50.

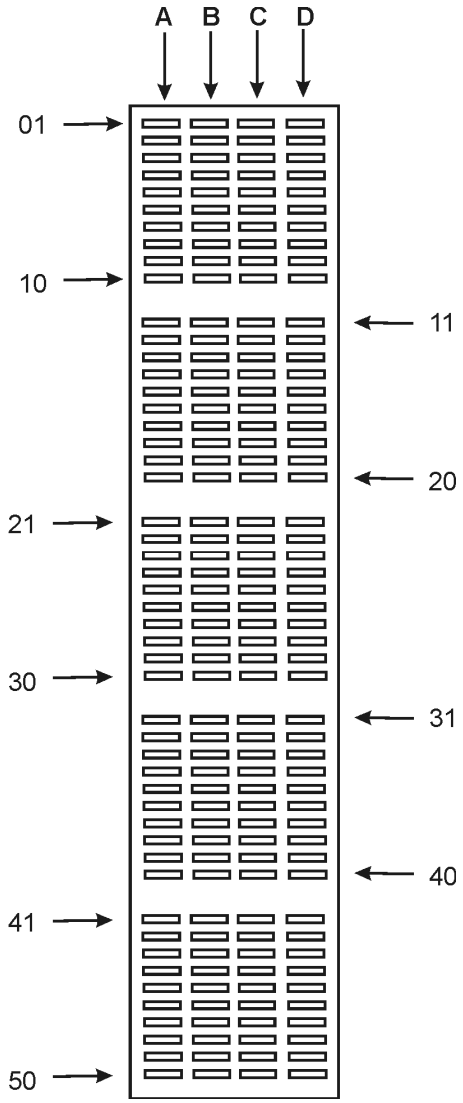


Figure 11.23
Punch down block map for UTP cabling

11.11.4 Connectors for fiber optic cables

Both multimode and single mode fiber optics use the same standard connector in the Ethernet 10BaseFL and FOIRL specifications.

Straight-tip (ST) connector

The 10BaseFL standard and FOIRL specification for Ethernet networks define one style of connector as being acceptable for both multimode and single mode fiber optic cabling – the straight-tip or ST connector (note that ST connectors for single mode and multimode fiber optics are different in construction and are not to be used interchangeably). Designed by AT&T, the ST connector replaces the earlier sub-miniature assembly or SMA connector. The ST connector is a keyed, locking connector that automatically aligns the center strands of the fiber optic cabling with the transmission or reception points of the network or cable management device it is connecting to.

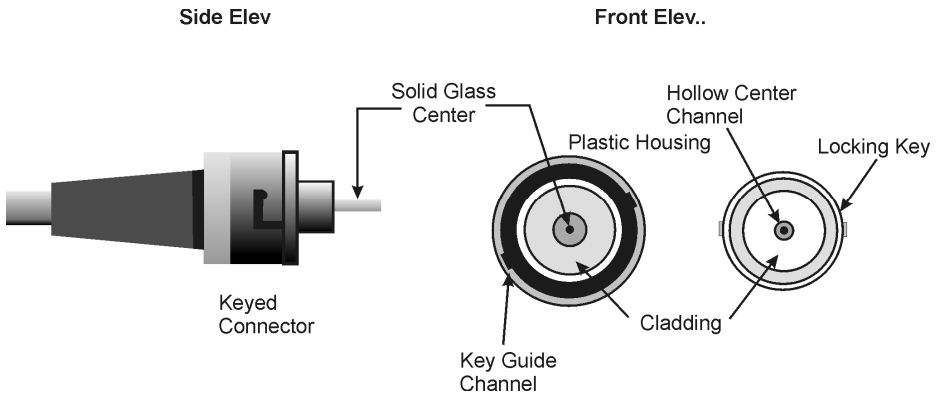


Figure 11.24
ST connector

The key guide channels of the male ST connector allow the ST connector to only be connected to a female ST connector in the proper alignment. The alignment keys of the female ST connector ensure the proper rotation of the connector and, at the end of the channel, lock the male ST connector into place at the correct attitude. An integral spring holds the ST connectors together and provides strain relief on the cables.

SC connector

The SC connector is a gendered connector that is recommended for use in Fast Ethernet networks that incorporate multimode fiber optics adhering to the 100BaseFX specification. It consists of two plastic housings, the outer and inner. The inner housing fits loosely into the outer, and slides back and forth with a travel of approximately 2 mm (0.08 in). The fiber is terminated inside a spring-loaded ceramic ferrule inside the inner housing. These ferrules are very similar to the floating ferrules used in the FDDI MIC connector.

The 100BaseFX specification requires very precise alignment of the fiber optic strands in order to make an acceptable connection. In order to accomplish this, SC connectors and ports each incorporate 'floating' ferrules that make the final connection between fibers. These floating ferrules are spring loaded to provide the correct mating tension. This arrangement allows the ferrules to move correctly when making a connection. This small amount of movement manages to accommodate the subtle differences in construction found from connector to connector and from port to port. The sides of the outer housing are open, allowing the inner housing to act as a latching mechanism when the connector is inserted properly in an SC port.

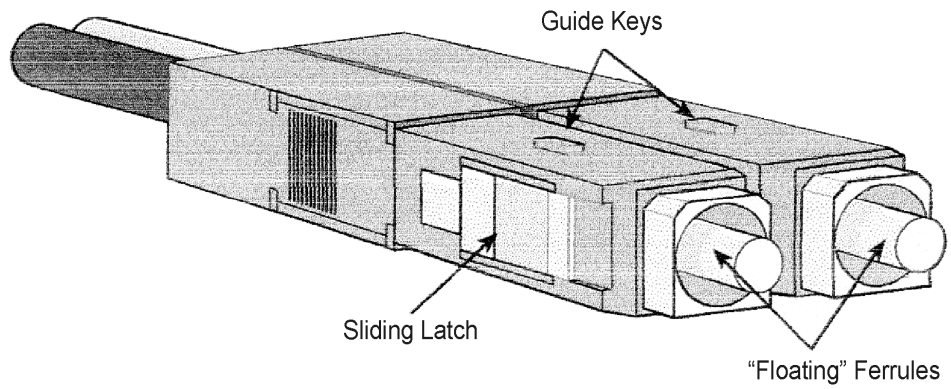


Figure 11.25
SC connectors (showing 2) for Fast Ethernet